Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/04/2007
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Project Title:	Mechanisms of Skeletal Muscle Atrophy Due to Simulated Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 6/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jackman, Robert (Boston University)		
Grant/Contract No.:	NNA04CD02G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	It is well known that exposure to spaceflight leads to marked decreases in skeletal muscle size and functional capacity, but the mechanisms underlying these changes are poorly understood. During the previous 3 yrs of NSBRI support, our lab has completed an extensive microarray analysis of soleus muscles over a time course of unloading. These data will be used as a source of candidate genes to test for regulatory roles in the atrophy process using genetic approaches in muscle cell culture and in whole muscle. The overall aim of this project is to identify genes involved in the regulation of protein loss during muscle atrophy. In addition we will test drugs of the NSAID family that have been suggested to have anti-atrophy effects. We will also test whether specific drugs that inhibit proteins involved in NF-kappaB signaling, which are known to be required for atrophy, can inhibit atrophy due to simulated weightlessness. These drugs include, helenalin, andrographolide, and wedelolactone. In work to identify genes that are sufficient to induce atrophy we will use genes from the microarray analysis that were activated early with unloading, had sustained activation, and are involved with protein degradation. These genes include cathepsins C, D, and L, the ubiquitin ligases Nedd4 and atrogin-1, and calpastatin. As atrophy also involves decreased protein synthesis, we will also study IGFBP5, a gene involved in the negative growth control of IGF-1. As endpoint markers of atrophy we will measure cell diameter and DNA/protein ratio, but we will also measure changes that represent intermediate steps in the atrophy process. These include the phosphorylation state of proteins involved in translation, the activity of proteolytic systems, and protein synthetic and degradation rates. In addition, co-expression strategies will be used to test whether multiple genes more effectively induce markers of atrophy. Where possible we will also test whether the inhibition of candidate genes can attenuate atrophy in a myotube atrophy model, thus testing whether the candidate is necessary for atrophy in cell culture. To test if a candidate gene is necessary for unloading induced atrophy, we will inhibit its expression by injecting a dominant negative or an anti-sense version of the candidate gene contained in plasmid or viral expression vectors. We will then measure if unloading atrophy can be attenuated in fibers containing the transgene. Uptake by individual muscle fibers will be tracked using epitope-tagged cDNAs or EGFP fusion proteins in order to colocalize overexpression with the effects on fiber size. Knockout mice will also be used to evalute the ability of candidate genes to attenuate unloading muscle atrophy. By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to exposure to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to any type of disuse atrophy.
Task Progress:	Construction & use of dominant negative NF-kappaB proteins as tools to learn about muscle atrophy We have shown that members of the NF-kappaB (NF-kB) signaling pathway are involved in disuse atrophy. We have already published data demonstrating that p50 and Bcl-3 are required (using knockout mice) for atrophy and we have also shown that IkB and IKKbeta also play a role. The NF-kB transcription factor c-Rel is not involved since knockout mice atrophied normally due to 10 days of disuse. To determine a role for additional NF-kB signaling proteins and to investigate in more detail the role of the proteins mentioned above additional genetic tools were made. In the last funding period EGFP fusion proteins were constructed for dominant negative (d.n.) IKKbeta, IKKalpha, IKKgamma (a.k.a. NEMO) and Rel A (p65). Overexpression of each of these dominant negative proteins was meant to determine if the normal endogenous function of the protein was required for atrophy. An EGFP fusion was made in order to visualize the fibers that expressed the dominant negative protein. After construction, each of these fusion proteins were tested in C2C12 cultured muscle cells to assess their appropriate expression and function. Each new fusion protein was overexpressed in C2C12 cells, and as expected, inhibited TNF-induced NF-kB transcriptional activity as determined by NF-kB-specific reporter assay. These constructs were then ready to use in whole rodent experiments. The study of TNF as an atrophy triggering agent in muscle cells Since TNF is a catabolic cytokine that strongly activates NF-kB signaling and is thought to trigger atrophy in many types of cachexia, we also investigated the role of TNF induced NF-kB signaling in muscle cell culture. Wild type p65 strongly activated kB reporter activity (40-fold) and TNF (10ng/ml) further increased the reporter activity (150-fold). Bcl-3 and p50 did not activate the co-transfected kB reporter when overexpressed individually, but when expressed together NF-kB activity increased by 2-fold. When TNF was added to the mix, NF-kB activity increased to levels not different from cells treated with TNF alone. Co-transfection of these 3 proteins however, showed the greatest to levels not different from kB activity (100-fold), even greater than overexpression of p65 alone. Addition of TNF increased NF-kB activity to 120-fold above background. Importantly though, basal NF-kB activity in bcl3-/- fibroblasts was barely above background. TNF treatment of these cells increased kB activity by 16-fold even though the raw luciferase value was still very low. Rescue of bcl-3 by overexpression recovered basal NF-kB activity and TNF-induced NF-kB activity increased. These data show that all 3 of these NF-kB factors are required for TNF induced NF-kB activity. Ongoing experiments are designed to further understand the NF-kB signaling proteins required for TNF induced muscle catabolism, including the target genes. In resting C2C12 myotubes, p65 primarily localized to the cytosol while d.n. p65 localized to the nucleus (and did not activate the kB reporter as mentioned above). Bcl-3 and p50 localize to the nucleus in C2C12 cells. Localization was possible because we also made EGFP fusion proteins of wild type p65, p50, and Bcl-3. Use of additional genetic tools to study the role of NF-kB proteins in atrophy In order to study in more detail the role of p65, p50, and Bcl-3 in muscle atrophy, we purchased several shRNA plasmids to knockdown the endogenous expression of these proteins. These plasmids were first tested in cell culture to determine if they indeed knocked down the expression of endogenous protein that they were meant to target. Since knockdown of any of these 3 proteins will decrease NF-kB activity, we used our kB reporter as a read-out to assess the efficacy of each shRNA. C2C12 cells are derived from mouse so shRNA plasmids specific for mouse were purchased from SuperArray Bioscience. An advantage of these plasmids is that they encode EGFP from a different Pol II promoter while a Pol III promoter drives expression of the shRNA. Thus one can visualize the cells transfected (this is most important in rodent whole muscle experiments). However, in all our experiments in C2C12 cells only the shRNA for p50 (i.e., p105, which is the unprocessed form of p50) induced a significant decrease in kB reporter activity (70% decrease). The reason that we use a protein readout to assess knockdown of these genes is that a decrease in mRNA is not useful to us unless the protein is also decreased. Therefore, only in the case of p50 will we use the shRNA plasmid in mouse disuse atrophy studies. We are now injecting

this plasmid in mouse muscle to knockdown endogenous expression of p50 and we will discover whether disuse atrophy is inhibited. This study is meant to complement the published p50 knockout study. We will need to identify different shRNA sequences that will knockdown endogenous p65 and Bcl-3.

The role of the cytoskeleton in TNF-induced NF-kB signaling in muscle cells In order to assess the role of NF-kB in muscle cell culture we set up an experimental system in which we transfected an NF-kB sensitive luciferase reporter plasmid into C2C12 myoblasts and then allowed the cultures to differentiate, for various times, before treating them with TNF-alpha. We first found that a range of TNF doses, from 2 to 20 ng/ml, produced similar results on NF-kB induction. Choosing two of the doses, 5 and 10 ng/ml, we next observed that in this system, the effect of TNF is to increase NF-kB expression with time throughout a 24 hour period. Using a dose of 10 ng/ml and a time of 3 hours as our assay we began to look for other compounds that could replace, add to or antagonize the effect of TNF on NF-kB. One of the intriguing results that we found is that colchicine inhibits TNF induction of NF-kB. Since we had shown that an inhibitor of microtubules could oppose the effects of TNF on NF-kB it was logical to look at the effect of taxol, a microtubule stabilizer. Taxol could act in the same way as TNF, to induce NF-kB reporter expression. As a control we demonstrated that taxol does not cause an increase of TNF in the muscle cells. When we investigated whether the effect of TNF and taxol were additive, we found that the sum of taxol and TNF was never a greater effect than that of TNF alone. We next went about looking for parts of the NF-kB signaling pathway that are either shared or distinct between TNF and taxol. We looked at several inhibitors of TNF NF-kB induction and found that those inhibitors were always inhibitory for taxol effects. This included antagonists of p65 (helenalin), p50 (andrographolide) and the IkappaB kinases (Wedelolactone).

We wondered whether the effect of microtubule effectors was specific to muscle cells and so we tested the same conditions on A9 cells, an L cell derivative that is used as a test system for the toxic effects of TNF. We found that A9 cells showed a diminished absolute effect of colchicine or taxol when compared to that in C2 cells but that the qualitative effects were the same, both the positive effect of taxol and the inhibitory effect of colchicine to both TNF and taxol were observed with the A9 cells.

Since microtubules were indicated to be linked to NF-kB activation by TNF in muscle cultures, we looked at inhibitors of microfilaments, the cytochalasins. The cytochalasins required much higher levels to achieve the same degree of inhibition as colchicine, however they too antagonized the NFkB induction of TNF. We also wondered if a more detailed time course experiment would show any differences between either microtubule or microfilament manipulation in C2 or A9 cells. Again, we found that the effects on C2 and A9 cells were qualitatively similar. There was a very similar dose requirement for inhibition of either microtubules or microfilaments on TNF mediated NF-kB induction. The microtubule system (inhibited by colchicine) appeared to be much more sensitive than the microfilament system. Recently, a paper by Fazal et al. has shown that at low levels, cytochalasin D is disruptive to NFkB induced by thrombin but not induced by TNF. Since we had observed cytochalasin D inhibition to TNF, we performed the experiments at lower levels and have replicated their results; at 5 uM cytochalasin D does not effect the NFkB response by TNF (our previous dose response curve began at almost 10 uM), while 5uM cyto D does show inhibition of thrombin induced NF-kB. However, we also find that colchicine at 5 uM inhibits both TNF and thrombin induced NF-kB. Perhaps microtubules are more central to the cytoskeleton basis of NF-kB function than microfilaments.

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